

Is Oblique Lateral Interbody Fusion a Key to Solve Degenerative Lumbar Scoliosis?

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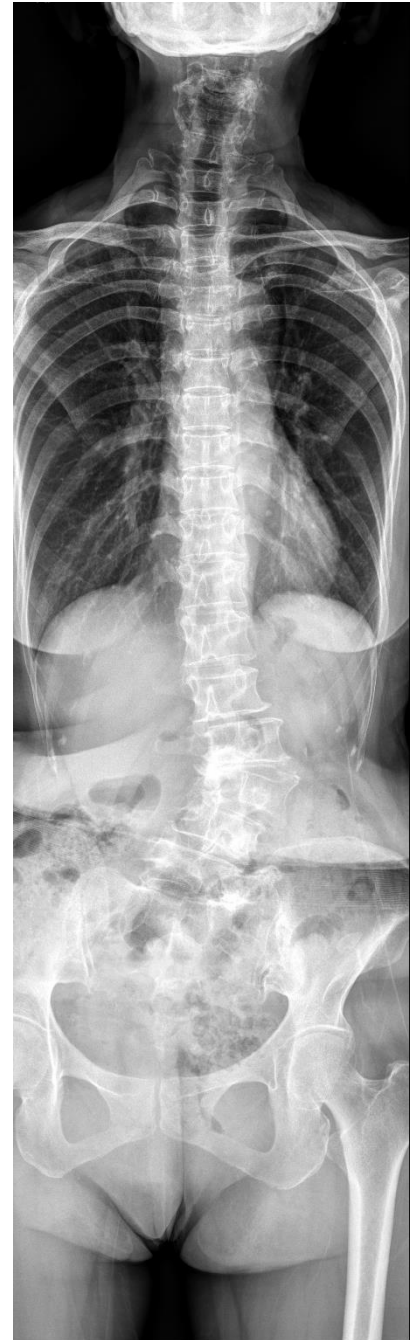
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Declarations

- **No funds or benefit were received from a commercial party.**
- **The authors have no potential conflicts of interest to disclose.**

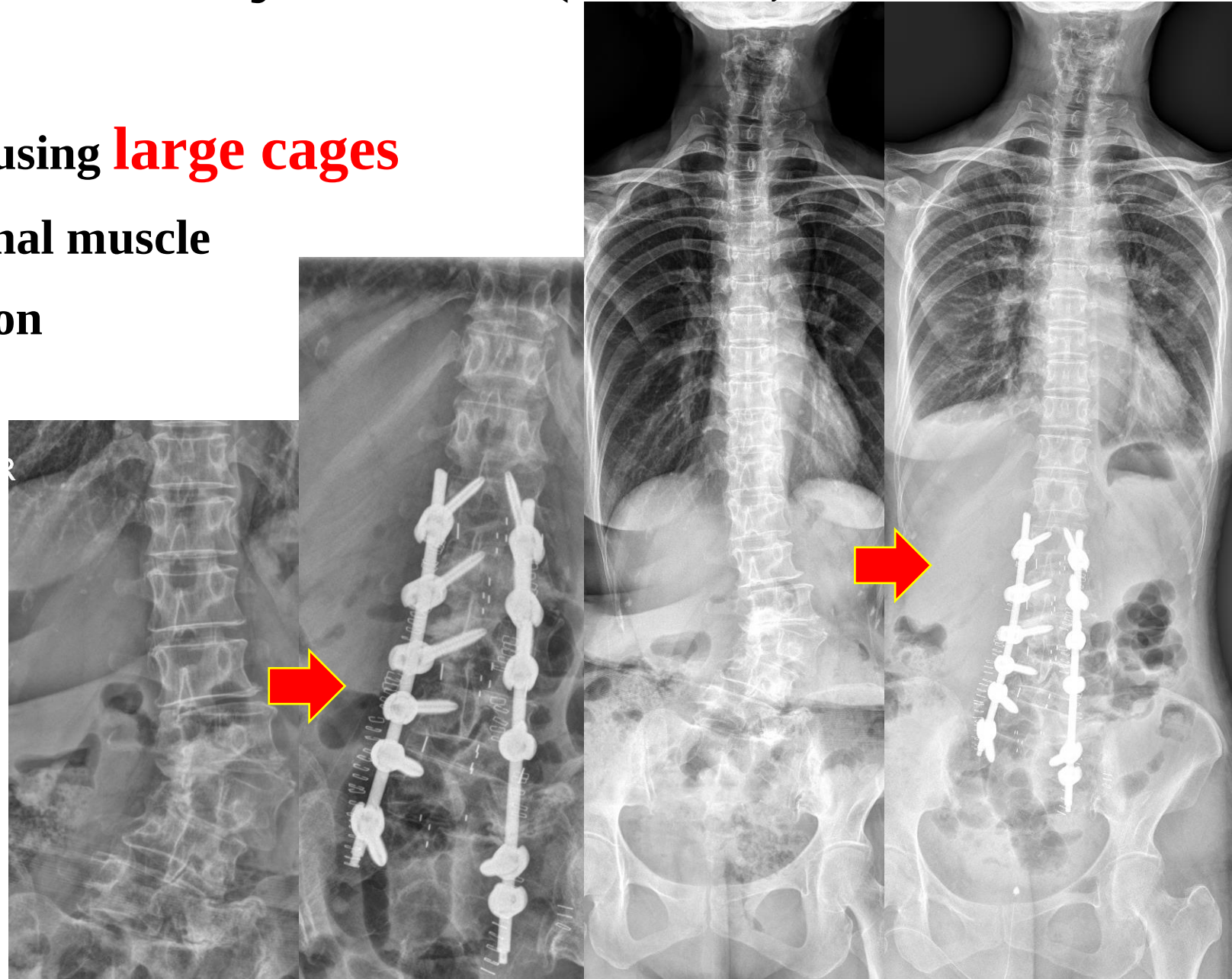
Introduction

- Degenerative lumbar scoliosis (**DLS**) is an abnormal curvature of the spine affecting any of the three planes: sagittal, axial, or coronal.
- De novo degenerative lumbar scoliosis that develops during adulthood is defined as a **coronal Cobb's angle** greater than **10 degrees**.
Silva F.e.et al., Neurosurg Focus, 2010.
- The most common complaints of patients seeking care for DLS are pain and disability. The majority of patients present with diffuse and axial low back pain.



Oblique Lateral interbody fusion (OLIF)

- Potential advantages
 - Good corrective power using **large cages**
 - Preservation of paraspinal muscle
 - No epidural manipulation



Purpose

- This study aims to compare the surgical and radiological variables of **OLIF** and **PLIF** in **DLS** patients.

Materials and Methods

▪ Inclusion criteria

- Patients with **DLS** who underwent **three or more levels of OLIF or PLIF** since September 2005.
- A preoperative coronal Cobb's angle greater than **10 degrees**.

▪ Exclusion criteria

- **Previous fusion surgery** in lumbar spine
- **Concurrent OLIF and PLIF/TLIF** performed at alternating levels.
- Malignancy, inflammatory disease, or infection.

Outcomes

▪ Demographics

- Age, sex, BMI
- Diagnosis
- Past-medical history
- 11-item modified frailty index (mFI-11)
- 5-item modified frailty index (mFI-5)

▪ Treatment-related data

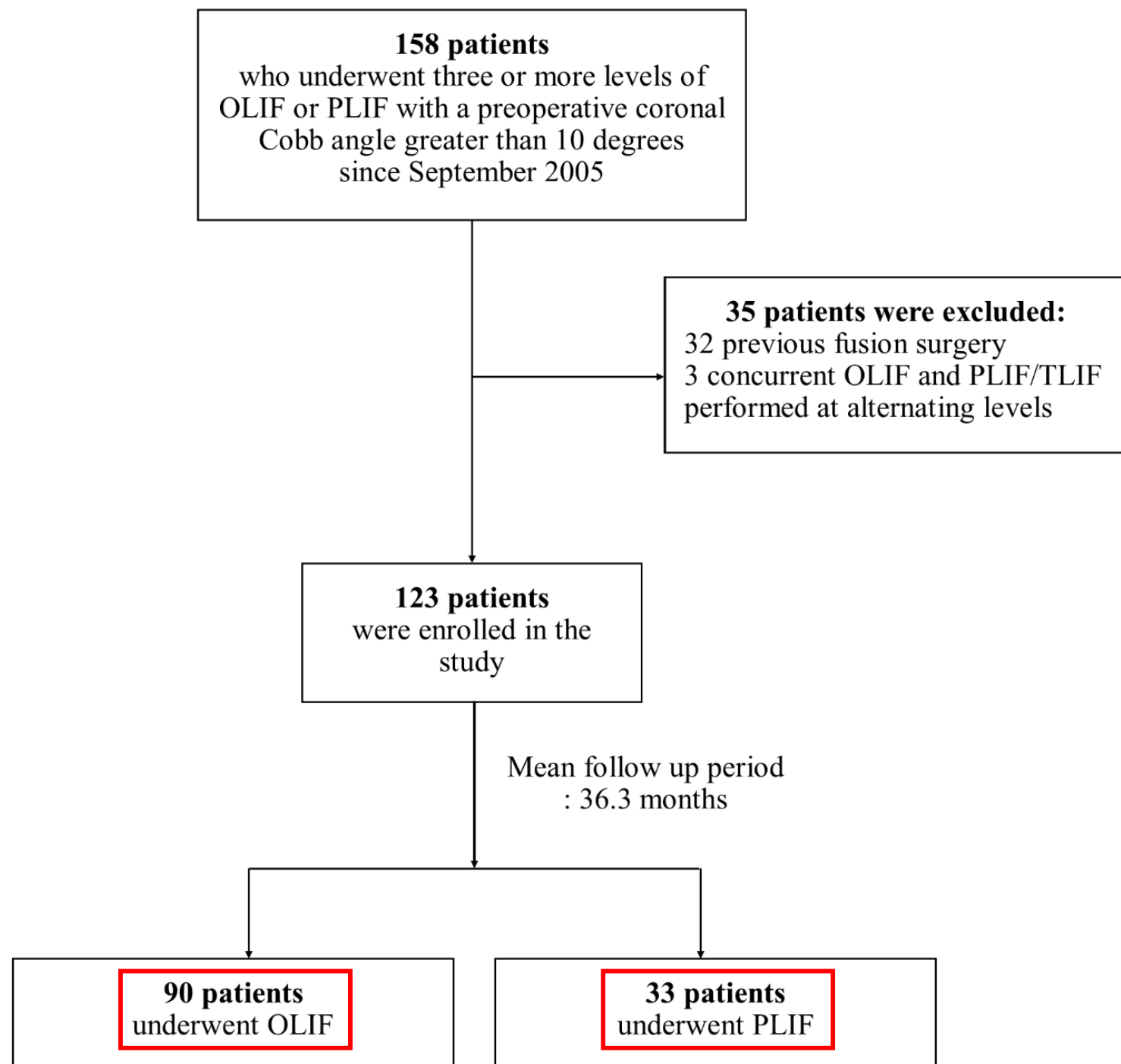
- no. of levels
- previous op. history
- operation time, anesthesia time
- Blood loss, RBC transfusion
- Amount of PCA usage
- Day of ambulation after surgery
- Hospital stay after surgery
- Complications

Outcomes

- Radiological factors: Preop. & postop. X-ray (also postoperative 1-year if exist)
 - Cobb's angle
 - C7-coronal vertical axis (CVA)

Results

- Study cohort -



Results

- Comparison of *baseline characteristics* -

Baseline characteristics	All patients		
	OLIF (n=90)	PLIF (n=33)	P-value
Age (mean $\pm$ SD, years)	70.3 $\pm$ 5.4	68.6 $\pm$ 5.1	0.116
Sex, Male (n [%])	17 [18.9]	6 [18.2]	0.929
BMI (mean $\pm$ SD, kg/m ²)	25.1 $\pm$ 3.3	25.5 $\pm$ 3.2	0.592
<u>Follow-up period (mean $\pm$ SD, months)</u>	30.7 $\pm$ 23.2	< 51.2 $\pm$ 36.7	0.005
mFI-11 (mean $\pm$ SD)	0.10 $\pm$ 0.07	0.10 $\pm$ 0.07	0.956
mFI-5 (mean $\pm$ SD)	0.19 $\pm$ 0.14	0.22 $\pm$ 0.16	0.409
Osteoporosis (n [%])	14 [16.9]	3 [14.3]	0.536
Number of interbody fusion (mean $\pm$ SD)	3.4 $\pm$ 0.9	3.3 $\pm$ 0.5	0.377
<u>Previous history of spine surgery (n [%])</u>	19 [21.1]	> 2 [6.1]	0.049
Preoperative main diagnosis (n [%])			
Spinal stenosis without SPLT	31 [34.4]	7 [21.2]	0.295
Degenerative SPLT	33 [36.7]	18 [54.5]	
Spondylolytic SPLT	1 [1.1]	0 [0.0]	
Degenerative lumbar deformity	25 [27.8]	8 [24.2]	

Results

- Comparison of *operative and postoperative variables* -

Variables	All patients			
	OLIF group (n=90)		PLIF group (n=33)	P-value
<u>Operation time</u> (mean ± SD, minute)	342.4 ± 91.4	>	250.4 ± 47.9	<0.001
<u>Anesthesia time</u> (mean ± SD, minute)	397.9 ± 96.9	>	303.5 ± 51.7	<0.001
<u>Estimated blood loss</u> (mean ± SD, ml)	853.8 ± 1503.9	<	1964.6 ± 1104.7	<0.001
<u>RBC transfusion</u> (mean ± SD, unit)	1.8 ± 3.4	<	5.6 ± 3.2	<0.001
Amount of PCA usage (mean ± SD, unit)	1.3 ± 0.6		1.2 ± 0.4	0.395
<u>Day of ambulation</u> after surgery				
(mean ± SD, day)	1.6 ± 1.0	<	3.1 ± 2.0	<0.001
<u>Hospital stay</u> after surgery				
(mean ± SD, day)	7.3 ± 3.0	<	9.7 ± 3.4	<0.001

Results

- Comparison of *radiologic parameters* -

Radiologic parameters	All patients		
	OLIF	PLIF	P-value
Preoperative Cobb angle	20.2 ± 8.0	18.6 ± 5.5	0.212
Postoperative Cobb angle	7.5 ± 4.0*	6.9 ± 4.6*	0.467
Δ Cobb angle	12.7 ± 6.5	11.7 ± 5.8	0.448
Preoperative C7-CVA	1.3 ± 22.8	-0.5 ± 26.9	0.736
Postoperative C7-CVA	-2.2 ± 19.5	0.7 ± 15.4	0.583
Δ C7-CVA	3.4 ± 21.3	4.9 ± 17.1	0.805
Cobb angle at postoperative 1-year	8.7 ± 4.8*	7.2 ± 5.5*	0.196
C7-CVA at postoperative 1-year	-0.7 ± 17.0	-3.9 ± 13.2	0.597

* $P < 0.05$ compared with preoperative.

Results

- Comparison of *complications*-

■ OLIF

- Proximal junctional kyphosis with compression fracture : 3 cases (3.3%)
- Adjacent segment disease : 1 case (1.1%)
- Infection : 5 cases (5.5%)
- Delayed union : 1 case (1.1%)
- Screw loosening : 4 cases (4.4%)
- Screw malposition : 2 cases (2.2%)

■ PLIF

- Proximal junctional kyphosis with compression fracture: 3 cases (9.1%)
- Adjacent segment disease : 3 cases (9.1%)
- Delayed union : 1 case (3.0%)
- Nonunion : 1 case (3.0%)
- Screw loosening : 1 case (3.0%)

Conclusions

- Both OLIF and PLIF effectively corrected the Cobb angle.
- **OLIF** entails **longer surgical and anesthesia durations** compared to PLIF. (can be overcome by single position)
- However, compared to PLIF, **OLIF** leads to **less blood loss** and **fewer transfusions**, and facilitates **quicker mobilization and discharge**.



Thank you for your attention!

